VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Fall Semester 2022-2023

Continuous Assessment Test -II (October 2022)

Slot: A1+TA1+TAA1 **Programme:** B.Tech(CSE/IT)

Course : BMAT205L – Discrete Mathematics and Graph Theory

Max. Time: 90 minutes **Max. Marks:** 50

Answer all the questions (5x10=50)

1. Verify Lagrange's theorem for the subgroup generated by 4 of the group $< Z_{17}^{*} , \times_{17} >$

2. An encoding function $e: B^3 \rightarrow B^6$ is given by the generator matrix $\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix}$.

   a) Determine all code words generated by the matrix. b) Find the associated parity check matrix H c) Use H to decode the following received words 100100, 010100

3. a) How many positive integers between 100 and 999 inclusive
   (i) are not divisible by either 3 or 4? (ii) are divisible by 3 but not by 4?

   b)How many different ways are there to choose 3 donuts from the 5 varieties at a donut shop? Assume that there are at least 3 donuts of each variety.

4. Solve the recurrence relation
   $a_{n+2} + 3a_{n+1} + 2a_n = 3^n, \ n \geq 0, a_0 = 0, a_1 = 1$

5. Draw the Hasse diagram for the poset $\{S_{30}, D\}$, divisors of 30 under the relation divides. Hence or otherwise prove that it is a lattice, also find $3 * (10 \oplus 3)$ and $3 \oplus (10 * 3)$

A1 CAT II DISCRETE

1 The subgp $H = \{1, 4, 13, 16\}$

Left cosets $1H$, $2H = \{2, 8, 9, 15\}$

$3H = \{3, 12, 5, 14\}$ $6H = \{6, 7, 10, 11\}$

$H \cup 2H \cup 3H \cup 6H = Z_{17}^* \; Z_{17}^*$

Also all cosets are disjoint.

$\therefore |H| + |2H| + |3H| + |6H| = |Z_{17}^*|$

i.e $4 \times 4 = 16$

$\Rightarrow 4 \mid 16$

2. $G = (I_m \mid A)$ $\therefore A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

$H = (A^T \mid I_{n-m}) = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$

$B^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$

$e(000) = (000)G = (000000)$

$e(001) = (001101)$ $e(101) = (101011)$

$e(010) = (010011)$ $e(110) = (110101)$

$e(011) = (011110)$ $e(111) = (111000)$

$e(100) = (100110)$

$H \cdot (100100)^T = (010)^T$ $H \cdot (010100)^T = (111)^T$

5th column of H

$\therefore$ decoded word 100110

& original msg is 100

not identical with any column of H

$\therefore$ can not be decoded.

3.a $|A| = \left\lfloor \frac{999}{3} \right\rfloor - \left\lfloor \frac{99}{3} \right\rfloor = 300$

$|B| = \left\lfloor \frac{999}{4} \right\rfloor - \left\lfloor \frac{99}{4} \right\rfloor = 225$

$|A \cap B| = \left\lfloor \frac{999}{12} \right\rfloor - \left\lfloor \frac{99}{12} \right\rfloor = 75$

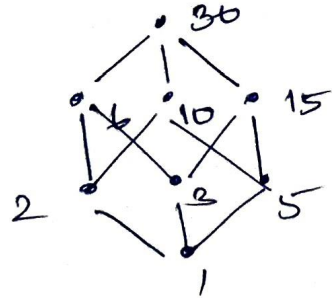
$\therefore |A \cup B| = 450$

i) $900 - 450 = 450$ ii) $300 - 75 = 225$

b) $C(5+3-1, 3) = 7C_3$ ways.

4. $a_n = \frac{-4}{5}(-2)^n + \frac{3}{4}(-1)^n + \frac{1}{20}(3^n)$

5.



$3 * (10 \oplus 3) = 3 * 30 = 3$

$3 \oplus (10 * 3) = 3 \oplus 1 = 3$